

Note that X is Panchishkin ordinary if there exist a Greenberg local condition Δ such that (X, Δ) is 0-Panchishkin and all Hodge–Tate weights of $\mathcal{F}_i^+ X$ (resp. of $X/\mathcal{F}_i^+ X$) are positive (resp. non-positive) for $i = 1, 2$.

Example 3.12. —

- (i) Let $x \in \mathcal{W}_{\text{cl}}^{(\text{f}, \text{bal})}(O)$ be an O -valued arithmetic point and let us denote $X(x) := X \otimes_x O$ for $X = \mathbf{T}_1, \mathbf{T}_2$, and $\mathbf{T}_3$. Then

$$\begin{aligned}\delta_{\text{Pan}}(\mathbf{T}_3(x), \Delta_{(\mathbf{g}, \text{bal})}) &= 0; \\ \delta_{\text{Pan}}(\mathbf{T}_2(x), \text{tr}^* \Delta_{(\mathbf{g}, \text{bal})}) &= 1 = \delta_{\text{Pan}}(\mathbf{T}_1(x), \text{tr}^* \Delta_{(\mathbf{g}, \text{bal})}).\end{aligned}$$

- (ii) Let $x \in \mathcal{W}_{\text{cl}}^{\text{f}}(O)$ be an O -valued arithmetic point and let us denote $X(x) := X \otimes_x O$ for $X = \mathbf{T}_1, \mathbf{T}_2$, and $\mathbf{T}_3$. Then

$$\begin{aligned}\delta_{\text{Pan}}(\mathbf{T}_3(x), \Delta_{\mathbf{g}}) &= 0 = \delta_{\text{Pan}}(\mathbf{T}_2(x), \Delta_{\text{bal}}); \\ \delta_{\text{Pan}}(\mathbf{T}_2(x), \text{tr}^* \Delta_{\mathbf{g}}) &= 2 = \delta_{\text{Pan}}(\mathbf{T}_1(x), \text{tr}^* \Delta_{\mathbf{g}}).\end{aligned}$$

Remark 3.13. Using the Panchishkin defect we can indicate why the factorization problem for algebraic p -adic L -functions we consider in this article is considerably more challenging than those problems considered earlier in [Gre82, Pal18, BCS23]. All the factorizations result in op. cite are obtained when the associated pairs are either 0-Panchishkin or 1-Panchishkin while we also consider the 2-Panchishkin pairs (cf. [BCS23, §4.3.2]). This discussion also gives evidence towards the existence of higher rank Euler systems assumed in Section 6 (cf. [LZ20b]).

3.3. Remarks on Tamagawa Factors. Let $v \nmid p\infty$ be a place in F . Suppose $\varphi: G_{F_v} \rightarrow \text{GL}_R(V)$ be a continuous G_{F_v} -representation, where V is a free module of finite rank over a complete Noetherian local ring R with finite residue field of characteristic p .

Remark 3.14. When $R = \mathbb{Z}_p$, $F = \mathbb{Q}$, the p -adic valuation of the order of $H^1(I_v, V)_{R\text{-tors}}^{\text{Fr}_v=1}$ is the local Tamagawa factor at v (see [FPR94, §I.4.2.2]). So it vanishes if $H^1(I_v, V)$ is a free R -module. In this philosophy, we would concentrate on the following conditions:

- The R -module $H^1(I_v, V)$ is free.

This discussion about the Tamagawa factor is needed for verification that our Selmer complexes are perfect.

3.3.1. Let $I_v^{(p)} \triangleleft I_v$ be the unique subgroup satisfying $I_v/I_v^{(p)} \cong \mathbb{Z}_p$. Let us fix a topological generator $t \in I_v/I_v^{(p)}$. (cf. [NSW08, §7.5.2])

Lemma 3.15. —

- (1) The R -module $V^{I_v^{(p)}}$ is free of finite rank.
- (2) The complex $C^\bullet(I_v, V)$ is quasi-isomorphic to the perfect complex

$$\cdots \rightarrow 0 \rightarrow V^{I_v^{(p)}} \xrightarrow{t-1} V^{I_v^{(p)}} \rightarrow 0 \rightarrow \cdots$$

concentrated in degrees 0 and 1.

- (3) For any ring homomorphism $R \rightarrow S$, the induced map $V^{I_v^{(p)}} \otimes_R S \rightarrow (V \otimes_R S)^{I_v^{(p)}}$ is an isomorphism. Therefore in the derived category

$$R\Gamma(I_v, V) \otimes_R^{\mathbb{L}} S \xrightarrow{\sim} R\Gamma(I_v, V \otimes_R S)$$

Proof. The statement (1) and (2) follows from [Nek06, §7.5.8].

To prove (3), [Nek06, §7.5.8] gives the inclusion $V^{I_v^{(p)}} \rightarrow V$ is split, the R -module $V/V^{I_v^{(p)}}$ is flat. hence we have an exact sequence of S -modules

$$0 \longrightarrow V^{I_v^{(p)}} \otimes_R S \longrightarrow V \otimes_R S \longrightarrow V/V^{I_v^{(p)}} \otimes_R S \longrightarrow 0.$$

Moreover, the group $I_v^{(p)}$ acts trivially on the ring S . Since $\varphi(I_v^{(p)})$ is finite and $p \nmid \#\varphi(I_v^{(p)})$, we have

$$e := \frac{1}{\#\varphi(I_v^{(p)})} \sum_{g \in \varphi(I_v^{(p)})} g \in O[\varphi(I_v^{(p)})]$$

Therefore we have

$$(V/V^{I_v^{(p)}} \otimes_R S)^{I_v^{(p)}} = e(V/V^{I_v^{(p)}} \otimes_R S) = e(V/V^{I_v^{(p)}}) \otimes_R S = 0.$$

This shows that the canonical homomorphism $V^{I_v^{(p)}} \otimes_R S \rightarrow (V \otimes_R S)^{I_v^{(p)}}$ is an isomorphism. $\square$

Corollary 3.16. Suppose $\varphi(I_v)$ is finite and $p \nmid \#\varphi(I_v)$ then R -module $H^1(I_v, V)$ is free.

Proof. Follows from Lemma 3.15 since in this scenario we have isomorphism of R -modules $H^1(I_v, V) \cong V^{I_v^{(p)}}$. $\square$